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Synopsis - Major Project

Project Title: PowerPulse: A Data-Driven Approach to Forecasting Electricity Demand

❖ Introduction

Electricity consumption forecasting is essential for efficient energy planning and sustainable power management. Accurate prediction of energy usage helps in reducing wastage, controlling costs, and ensuring proper resource allocation.

This project aims to analyze historical electricity consumption data and develop a time-series forecasting model to predict future energy demand. An interactive dashboard will be developed to visualize trends and forecasted values.

❖ Problem Statement

Electricity consumption varies due to seasonal changes, human activities, and other external factors. Traditional estimation methods often fail to accurately capture these patterns.

There is a need for a data-driven forecasting system that can predict future electricity usage based on historical consumption data.

❖ Objectives

- To analyze historical electricity consumption data.
- To implement time-series forecasting models such as ARIMA and Prophet.
- To evaluate model performance using MAE and RMSE.
- To forecast electricity consumption for upcoming periods (~ 30-90 days).
- To develop an interactive dashboard for visualizing historical trends and predictions.

❖ Methodology

Step 1: Data Collection - Data Collection from public datasets.

Step 2: Data Preprocessing - (cleaning, handling missing values, formatting date-time)

Step 3: Exploratory Data Analysis (EDA) - to identify trends and seasonality

Step 4: Model Development - using ARIMA/Prophet

Step 5: Model Evaluation - using performance metrics

Step 6: Forecasting - Forecast Generation for future periods

Step 7: Dashboard Development - using Power BI/Streamlit

❖ **System Architecture**

Data Source → Data Preprocessing → EDA → Model Training → Model Evaluation → Forecast Generation → Dashboard Visualization

❖ **Software / Hardware Requirements**

Software:

- Python (3.x)
- Jupyter Notebook / VS Code
- Pandas, Numpy, Prophet, Statsmodels
- Power BI / Streamlit

Hardware:

- Minimum 8GB RAM
- Intel i5 or equivalent processor

❖ **Expected Outcomes**

- Forecasted electricity consumption values.
- Identification of seasonal patterns and trends.
- Detection of anomalies in energy usage
- Interactive dashboard for visualization.

❖ **Future Scope**

- Integration with real-time smart meter data.
- Implementation of advanced deep learning models like LSTM.
- Deployment as a web-based application.